

Groundwater Extraction and the Unpriced Recharge Subsidy in Central Maui

Sean Barnes

BUS-620

Shidler College of Business

October 9, 2026

The Challenge

Central Maui's Kahului and Pā'ia aquifer systems are being pumped above their published sustainable-yield figures. The latest 12-month moving average through May 31, 2026 reports about 7.760 mgd from Kahului and 15.175 mgd from Pā'ia (Maui Now 2026) — against sustainable yields of 1 mgd and 7 mgd, respectively (CWRM 2019, Table F-11), or roughly 776% and 217% of yield. The same aquifer complex supplies Maui County's municipal system and the growth projected through 2035 (Maui County DWS 2026; 2025). Neither system is designated a Ground Water Management Area, so CWRM cannot yet impose a volume cap through water-use permits (Haw. Rev. Stat. §174C-41). Extraction is unpriced and unmeasured against a current recharge benchmark, while the imported water that may be providing that recharge is itself reduced whenever drought cuts East Maui deliveries.

What the Sustainable-Yield Numbers Measure

A sustainable yield is a planning benchmark, not a real-time account of every gallon. CWRM's Table F-11 note is central: published yields for Kahului and Pā'ia ignore significant return irrigation (water that seeps back and is later withdrawn unmetered). The 1 mgd and 7 mgd figures are primary CWRM values (CWRM 2019); the 7.760 mgd and 15.175 mgd figures are a moving average from Maui Now (2026) — reported, not primary, data.

The numbers also do not fully reconcile across time. The USGS analysis attached to CWRM's May 2026 staff submittal reports the full Central sector — Kahului, Pā'ia, Makawao, and Kama'ole — withdrew 16.7 mgd in 2023, with Kahului alone at 5.2 mgd (USGS 2025, Exhibit 2 to CWRM 2026). The May 2026 average above covers only two of those four systems and already totals about 22.9 mgd. Scope explains part of the gap, but not growth of that size in under three years — itself evidence the measurement is not in place.

The Economics

Four mechanisms explain why the system can overpump despite physical uncertainty. First, groundwater is a common-pool resource: Mahi Pono, Maui County's DWS, and other well users draw from the same aquifer (Maui County DWS 2026), and no single user can protect the shared lens alone. A commons is not a failure of the people in it but a right nobody assigned: with no owner there is nobody to bargain with, so the Coasean solution never starts. Hawai'i has assigned the right — the State holds all water in trust, and there are no vested rights in the use of public water (Waiāhole, 94 Haw. 97 (2000); Haw. Rev. Stat. § 174C-2(b)). However, a right is well defined only when it is enforceable, and without designation CWRM cannot enforce it. Private bargaining cannot substitute: the parties are too many and too diffuse.

Second, extraction is a negative externality: pumping that lowers water levels or pulls the transition zone inland costs municipal customers and the public, though the record does not yet prove measurable chloride damage here (USGS 2025). Third, groundwater is mispriced: a large user pays to operate a pump but not the scarcity cost imposed on future users. Fourth, return irrigation is a positive externality: it replenishes groundwater at no cost to the pumper (CWRM 2019), and like any uncompensated benefit it is under-supplied — nobody is paid to keep it flowing, and it shrinks whenever drought cuts East Maui deliveries.

Implications — What Happens Next

Central Maui pumping reached a much higher reported peak of about 120 mgd in 1996, before sugarcane ended (CWRM 2026). The issue is not that today's rate is the highest ever, but that pumping is high against published yield while the return-flow buffer contracts. No deep monitor wells exist in the study area to track the freshwater lens or transition zone (USGS 2025), and USGS ranks all four Central-sector systems high priority for monitoring (USGS 2020, quoted in CWRM

2026) — a documented monitoring gap, not a demonstrated harm, and the one this paper asks CWRM to close.

Recommendation

I recommend CWRM designate the Kahului and Pā‘ia aquifer systems as Ground Water Management Areas under Haw. Rev. Stat. § 174C-41, coordinating with the Central aquifer sector where flows connect them. The system is the right unit because CWRM sets and measures yields at that scale: regulating the sector instead would let Kama‘ole's unused allowance offset Kahului's deficit on paper without moving any water to Kahului's wells. Flow crossing the boundaries is a reason to model recharge at the sector scale, not to license at it. Kahului alone clears the statutory 90%-of-sustainable-yield designation trigger (§ 174C-44) on every reported vintage — 520–776% of its 1 mgd yield is not a marginal case.

CWRM should require large agricultural, commercial, and industrial users pumping over 100,000 gallons per day (0.1 mgd) to install audited digital meters — my proposed line, not an existing exemption. Year one establishes a verified baseline of withdrawals, deliveries, return flow, water levels and chloride; CWRM then sets a basin cap from updated sustainable-yield estimates. The fee applies above each large user's allotment and rises toward the cap — the schedule left unset until the baseline year produces the withdrawal and recharge figures a defensible rate must be built from. CWRM administers permits, audits records and penalizes tampering; agricultural users get one year to comply.

Among the standard instruments — tax, cap, permit, rule — the choice is settled by information, not ideology. A corrective tax must be set at the external cost — the number this basin cannot produce, since the yields exclude return irrigation and no monitor well measures the lens. A cap asks less, requiring a defensible view of the safe quantity, which permit reporting is designed to

generate. Demand here is weakly price-responsive besides: when East Maui deliveries fall, agriculture substitutes groundwater for want of a near-term alternative, so a fee alone would transfer revenue more than it would cut withdrawals. The fee funds monitoring and rations marginal use once the cap binds.

Existing operations would not be grandfathered: allotments follow from the year-one baseline and the updated yield rather than from historical use, since grandfathering a permit hands incumbents an asset. Residential and small-business users remain permanently exempt. Designation is justified not because the existing percentages are exact, but because the State cannot responsibly manage a recharge subsidy it does not measure or regulate.

Objections

The strongest objection is that overdraft appears overstated because imported surface water returns to the aquifer (CWRM 2019), and a cap based on the old denominator could restrict agriculture unnecessarily. My response: this argues for better measurement, not an unregulated basin — designation lets CWRM update the denominator and adjust the cap.

A second objection is that CWRM funded a roughly \$442,000 USGS study in May 2026 and should wait for it (CWRM 2026). The study answers a different question on a three-year horizon; year one of designation is itself a measurement period, and the cap can be revised as better estimates arrive.

Figure 1

Central Aquifer Sector: Withdrawals by System vs. Each System's Own Sustainable Yield

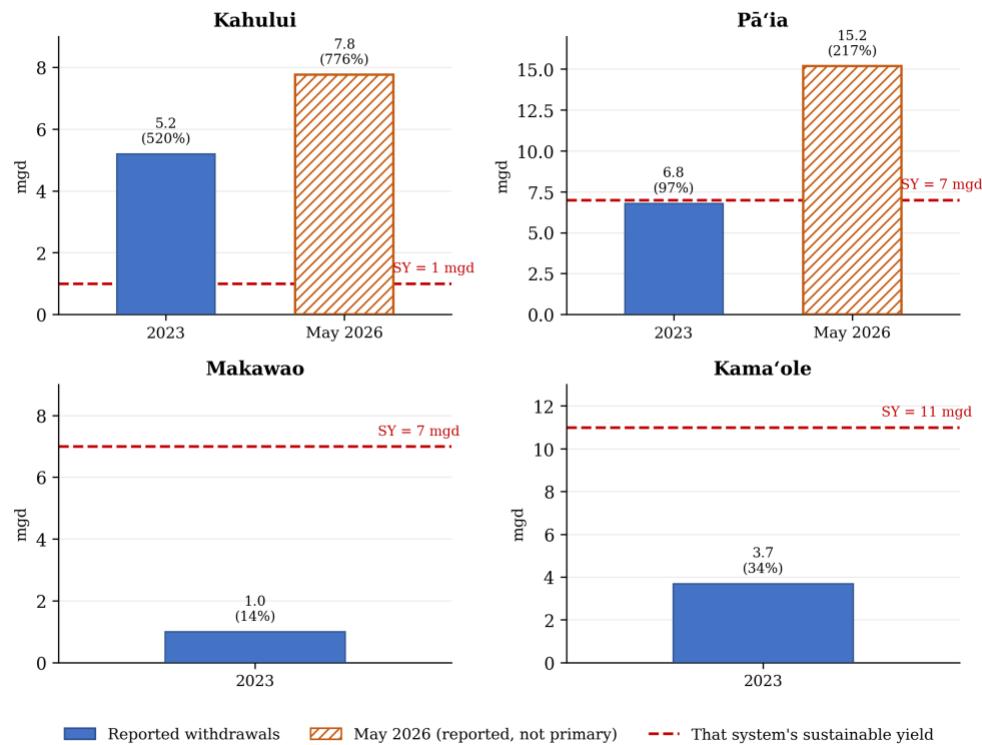


Figure 1. Reported withdrawals by aquifer system against each system's own CWRM sustainable-yield benchmark (CWRM 2019, Table F-11; USGS 2025; Maui Now 2026). Kahului stands at 520–776% of its 1 mgd yield across every reported vintage; Pā'ia is near its 7 mgd yield in 2023 and over it by May 2026; Makawao and Kama'ole sit well under their own yields (Kama'ole's 2023 figure is computed as the Central-sector remainder). May 2026 bars (hatched) cover only Kahului and Pā'ia and are reported, not primary, data.

Figure 2

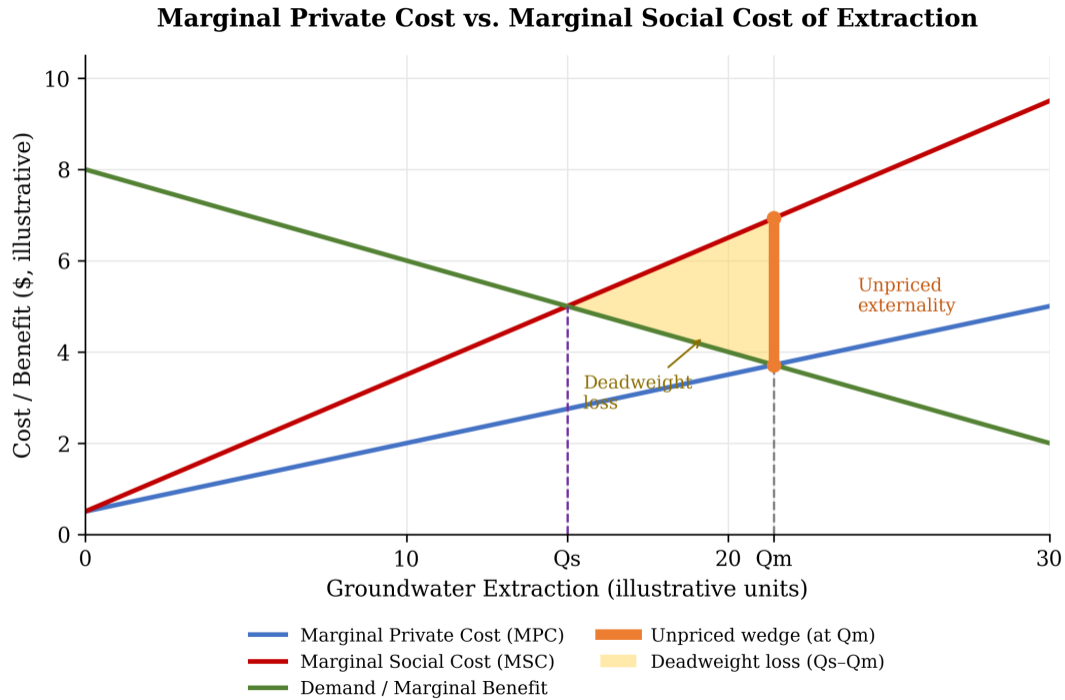


Figure 2. Illustrative diagram of the mechanism described in “The Economics”: because pumping is unpriced, users extract to Q_m , where demand meets only the marginal private cost (MPC), rather than to the efficient quantity Q_s , where demand meets the marginal social cost (MSC). The orange bracket marks the unpriced externality at Q_m ; the shaded triangle is the resulting deadweight loss. Curve shapes are illustrative and not fitted to measured cost data, since none exists; the downward-sloping demand curve is a simplification, for the reason the Recommendation gives.

References

- Commission on Water Resource Management. 2019. Water Resource Protection Plan 2019 Update. Honolulu: State of Hawai‘i Department of Land and Natural Resources. Table F-11, aquifer notes for the Kahului and Pā‘ia aquifer systems.
- Commission on Water Resource Management. 2026. “Staff Submittal B2: Joint Funding Agreement with the U.S. Geological Survey for ‘Analysis of Groundwater Flow in the Central Aquifer Sector, Maui, Hawai‘i.’” May 19, 2026.

Maui County Department of Water Supply. 2025. Water Use and Development Plan: Summary Document. February 2025.

Maui County Department of Water Supply. 2026. Water Source Report. February 10, 2026.

Maui Now. 2026. “Maui Planning Commission Recommends EC Pā‘ia Town Project Rezoning as Private Water Source Raises Questions.” September 10, 2026.

U.S. Geological Survey. 2020. Hydrological Monitoring Needs Assessment for the State of Hawai‘i. Not consulted directly; cited as quoted in CWRM 2026.

U.S. Geological Survey, Pacific Islands Water Science Center. 2025. “Analysis of Groundwater Flow in the Central Aquifer Sector, Maui, Hawai‘i.” January 2025. Exhibit 2 to CWRM 2026.

Legal authorities: In re Water Use Permit Applications (Waiāhole), 94 Haw. 97, 9 P.3d 409 (2000); Haw. Rev. Stat. §§ 174C-2(b), 174C-41 (designation of water management areas), 174C-44 (criteria, including the ninety per cent of sustainable yield threshold).